

OFX: Innovating the global money transfer industry with smart analytics and AI to prevent financial crime

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About OFX

OFX advanced its financial transparency working with Quantexa to implement an advanced contextual monitoring solution powered by Quantexa's CDI platform.

OFX is an Australian foreign exchange and payments company on a mission to promote fairer financial services through transparent, low-cost global money transfers. It serves more than 1 million customers, sending money to 190 countries.

Industries: Financial Services & Insurance

and deployed on
Cloud.

Location: Australia

Google Cloud results

- Facilitates smooth, transparent money transfers for OFX customers during 50% surge in global transactions with efficient financial crime detection
- Supports stringent financial-crime hunting by freeing up investigative team to probe serious

Smart
financial
alerts
cut in
crunc

quantexa

About Quantexa

Quantexa is a pioneer in the big data strategy known as “contextual decision intelligence.” Its software enables businesses to improve decision-making through a contextual view of people, places, and organizations. It is a Google Cloud financial services industry partner.

strategy known as “contextual decision
Google Cloud (<https://cloud.google.com/>)

Cloud SQL (<https://cloud.google.com/sql/>) ing
through a contextual view of people

Cloud Dataproc
(<https://cloud.google.com/dataproc/>)
Cloud financial services industry partner.

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Cloud Storage (<https://cloud.google.com/storage/>)

cases by
weeding out
irrelevant
alerts

- Inspires
creative
partnerships
on Google
Cloud
infrastructure
to evolve
detection
technology
that
promotes
financial
system
transparency

When OFX (<https://www.ofx.com/>) launched in 1998, its founders embraced a mission to promote a "better and fairer life" for people moving money around the world. The online money transfer platform enables everyone from large corporations to small businesses and individuals supporting their global lives sending money home, funding education, and more.

There's another key dimension to the OFX mission. "Embedded in our DNA," says COO Mark Shaw, "is investment in the fight against financial crime. Given the nature of our business, it's our responsibility to help to detect and report financial crime."

OFX handles thousands of transfers every day. And with volumes rising fast, OFX wanted a smarter monitoring system than the traditional "rules-based" industry standard, which sends financial crime alerts based on a static list of rules, such as unusually high transfer numbers in one week.

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—Mark Shaw, COO, OFX

In 2019, Mark discovered [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>) financial services industry partner [Quantexa](https://www.quantexa.com/) (<https://www.quantexa.com/>), and found the AI-driven solution he was looking for. Instead of a rules-based approach, Quantexa's system, powered by [Google Kubernetes Engine](https://kubernetes.io/) (/kubernetes-engine) (GKE) and tools such as [Dataproc](https://www.dataproc.com/) (/dataproc), offered an intelligent "model-based" system that probes networks and behavior patterns to uncover financial crime. It delivered superior quality, crunching the same amount of data in half the run time, for only a 5% rise in infrastructure costs.

"We continuously look for innovative ways to build on our existing prevention systems to keep ahead of financial crime," says Mark. "This matters because it protects our customers and our communities. Illicit flows can underpin horrendous crimes, and all of us in the industry must play our part in making sure we do everything in our power to ensure we prevent it. Quantexa's open technology deployed on Google Cloud integrates into our IT ecosystem and augments our team's investigative capability. It has given us the ability to identify wider networks of suspicious transfers, enabling us to react quickly and protect our customers."

Enabling smarter financial decisions with AI solutions powered by Google Cloud

Like the hard-boiled detective in a crime novel, OFX starts by identifying key players in the story and their relationships. It does this by using Quantexa's AI models powered by Google Cloud. The "players" are millions of data points in a global community of financial transaction. The first job of the AI "detective" is entity resolution, which refers to matching an anonymous data point (a number, string, or date) to a person, business, location, or anything else that exists in the physical world.

Step two is network analysis. This involves deploying AI algorithms to uncover relationships between those real-life "entities" that have been matched to the data. Strong connections (such as two people who shared an address at one point) are prioritized over weak connections (such as people who shared an address at different times, so probably don't know each other).

Deeper insights are uncovered when AI modeling reveals how strongly-connected entities have interacted over time. Like the detective with a mental database of criminal typologies, OFX and Quantexa have developed a library of networked behavioral patterns that, to a high degree of precision, indicate when a financial crime has been committed.

It's called contextual decision intelligence

(<https://www.acfcs.org/webinars/contextual-decision-intelligence/>)

(CDI), a new approach that unlocks the power of data and analytics across the enterprise. As the name suggests, CDI crunches data to generate a contextual picture of relationships that enables users to improve decision-making in any complex field, from global markets to government intelligence.

"We deploy
Elasticsearch in
Google
Kubernetes
Engine, which
gives us the
ability to scale
as required on a
granular process
level. And
because Google
is the architect
of Kubernetes,
we know that we
always have the
latest updates

and features
available with
Google
Kubernetes
Engine, ahead of
any other
provider."

—*Jamie Hutton, CTO,
Quantexa*

High-speed financial crime detection with Elasticsearch and Google Kubernetes Engine

Quantexa supports OFX by generating fewer alerts rather than more. The beauty of the system lies in providing quality intelligence that OFX's financial crime investigation team scrutinizes to get to the bottom of a case.

Google Cloud is key to Quantexa's ability to enable real-time AI-driven financial crime alerts, according to Quantexa CTO Jamie Hutton. In particular, Google Kubernetes Engine (GKE) provides Quantexa the scaling power to effectively deploy Elasticsearch ([https://www.elastic.co/elasticsearch/?ultron=\[EL\]-\[B\]-\[Stack\]-\[Trials\]-\[EMEA-C\]-Exact&blade=adwords-s&hulk=cpc&gambit=Elasticsearch-Core&Device=c&gclid=EAlaIqobChMI1KL14Jfr6gIVE-3tCh3U-gHOEAAYASAAEgLDXvD_BwE](https://www.elastic.co/elasticsearch/?ultron=[EL]-[B]-[Stack]-[Trials]-[EMEA-C]-Exact&blade=adwords-s&hulk=cpc&gambit=Elasticsearch-Core&Device=c&gclid=EAlaIqobChMI1KL14Jfr6gIVE-3tCh3U-gHOEAAYASAAEgLDXvD_BwE)), an enterprise search engine for data querying that's

central to Quantexa's entity resolution and network analytics.

"We deploy Elasticsearch in Google Kubernetes Engine, which gives us the ability to scale as required on a granular process level," says Jamie.

"And because Google is the architect of Kubernetes, we know that we always have the latest updates and features available with Google Kubernetes Engine, ahead of any other provider."

The superior financial crime-busting capabilities OFX gains from Quantexa also depend upon Google Cloud backend solutions, such as Cloud Storage (/storage), Cloud SQL (/sql), and Dataproc. Jamie called Dataproc "fantastic" for Quantexa batch processing on Apache Spark (<https://spark.apache.org/>).

High-speed data crunching combines with precision alerts to enable OFX to manage a continuous rise in transaction volumes. It means OFX can scale as fast as the demand requires and still improve its crime prevention. A key test came during the first peak of COVID-19 in March 2020, when OFX transaction volumes reached record levels.

"With the coronavirus, we saw some of our largest trading volume days in the company's history," says Mark. "Had we been on our existing system, we would have seen massive increases in the number of alerts to deal with, but with Quantexa, we kept everything under control by triaging quality alerts. It enabled people to move money quickly and confidently in a time of crisis, knowing we're working

to keep transactions safe from crime." Robust Google Cloud security on Quantexa's CDI infrastructure also helps to enable safe fund transfers for OFX customers.

Without divulging sensitive criminal cases, OFX shares one example with regulators and banking partners to demonstrate the power of the system. A customer set up a personal account and began making transactions with different entities around the world. CDI analysis of the customer's transactions made it clear they were in fact a corporate account set up as an individual. Using Quantexa's entity resolution and network visualization, investigators were able to track down the payment networks (its global web of inbound and outbound transfers) that indicated it was some kind of travel agency.

"While none of the individual transactions were particularly suspicious, the CDI system did flag to us that we had a customer that had started transacting on behalf of their business through their individual account," says Mark. "We were able to then work with that customer to onboard their business and ensure all the transactions were legitimate."

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—Mark Shaw, COO, OFX

Partnering for a future of financial inclusion and transparent global money flows

According to the Alliance for Financial Inclusion, fast and fair cross-border money flows are a powerful force in leveling the global playing field

(https://www.afi-global.org/sites/default/files/publications/2018-09/AFI_DFS_cross%20border_AW_digital.pdf)

. As OFX continues to make fair financial services a

reality for more customers, it's also busy creating new monitoring systems on Google Cloud with Quantexa, with the intent of staying two steps ahead of financial crime.

"We're really proud of the work with Quantexa and Google Cloud. It's much more than a basic vendor-client relationship, which is crucial when it comes to financial crime," says Mark. "Together, we've built a solution that empowers our people to help detect and prevent it."